

0.1 Riemann-Stieltjes Integral

Definition 0.1.1. A finite subset $P \subset [a, b]$ containing a, b is called a *Partition* of $[a, b]$. Conventionally, denote $P = \{x_0, x_1, \dots, x_n\}$ where $a = x_0 < x_1 < \dots < x_n = b$. The set of Partitions of $[a, b]$ is denoted as $\mathcal{P}[a, b]$.

Definition 0.1.2. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. Let a partition $P \in \mathcal{P}[a, b]$ be given, and denote $P = \{x_0, x_1, \dots, x_n\}$. For each $1 \leq i \leq n$, define

$$\Delta\alpha_i \stackrel{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\geq} 0$$

and

$$M_i \stackrel{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \leq t \leq x_i\} \text{ and } m_i \stackrel{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \leq t \leq x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n M_i \cdot [\alpha(x_i) - \alpha(x_{i-1})] \text{ and } L(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n m_i \cdot [\alpha(x_i) - \alpha(x_{i-1})]$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \inf_{P \in \mathcal{P}[a, b]} U(f, P, \alpha) \text{ and } \underline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \sup_{P \in \mathcal{P}[a, b]} L(f, P, \alpha)$$

We said that f is *Riemann-Stieltjes Integrable* respect to α if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions respect to α as $\mathcal{R}(\alpha)$.

Lemma 0.1.3. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. For any partitions $P, Q \subset [a, b]$, and a refinement $P^* \supset P$,

$$\underline{\int_a^b} f d\alpha \leq L(f, P, \alpha) \leq L(f, P^*, \alpha) \leq U(f, P^*, \alpha) \leq U(f, P, \alpha) \leq \overline{\int_a^b} f d\alpha$$

and

$$L(f, P, \alpha) \leq U(f, Q, \alpha)$$

Lemma 0.1.4. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. If $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, then $f + g, cf \in \mathcal{R}(\alpha)$ with

$$\begin{aligned}\int_a^b (f + g) d\alpha &= \int_a^b f d\alpha + \int_a^b g d\alpha \\ \int_a^b (cf) d\alpha &= c \int_a^b f d\alpha\end{aligned}$$

Lemma 0.1.5. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. If f is Riemann-Stieltjes Integrable respect to α on $[a, b]$ and $a < c < b$, then f is Riemann-Stieltjes Integrable respect to $[a, c]$ and $[c, b]$ with

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

Lemma 0.1.6. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha, \beta : [a, b] \rightarrow \mathbb{R}$ be monotone-increasing. If $f \in \mathcal{R}(\alpha)$, $f \in \mathcal{R}(\beta)$ and $c > 0$, then $f \in \mathcal{R}(\alpha + \beta)$ and $f \in \mathcal{R}(c\alpha)$ with

$$\begin{aligned}\int_a^b f d(\alpha + \beta) &= \int_a^b f d\alpha + \int_a^b f d\beta \\ \int_a^b f d(c\alpha) &= c \int_a^b f d\alpha\end{aligned}$$

Theorem 0.1.7. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be monotone-increasing. TFAE:

1. f is Riemann-Stieltjes Integrable respect to α on $[a, b]$.
2. For any $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a, b]$ such that $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$.
3. For some $A \in \mathbb{R}$, for any $\varepsilon > 0$, there exists $P_0 \in \mathcal{P}[a, b]$ such that

$$\forall P \in \mathcal{P}[a, b], P \supseteq P_0 \implies |S(f, P, \alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum $S(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$, for some $t_i \in [x_{i-1}, x_i]$.

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Definition 0.1.8. Let $\mathcal{P}[a, b]$ be a set of partitions. Define a *Norm* as:

$$\|\cdot\| : \mathcal{P}[a, b] \rightarrow \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1, 2, \dots, n\} \text{ where } P = \{x_0, x_1, \dots, x_n\}$$